

## CHAPTER III.5, WEEK 12

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5.1. Let  $X$  be a projective scheme over a field  $k$ , and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . We define the *Euler characteristic* of  $\mathcal{F}$  by

$$\chi(\mathcal{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathcal{F}).$$

If

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$$

is a short exact sequence of coherent sheaves on  $X$ , show that  $\chi(\mathcal{F}) = \chi(\mathcal{F}') + \chi(\mathcal{F}'')$ .

*Proof.* First, for an exact sequence of vector spaces  $0 \rightarrow V_0 \xrightarrow{f_0} \dots \xrightarrow{f_{n-1}} V_n \xrightarrow{f_n} 0$ , consider  $\sum_i (-1)^i \dim V_i$ . We have

$$\begin{aligned} \sum_{i=0}^n (-1)^i \dim V_i &= \sum_{i=1}^n (-1)^i \dim \ker f_i + \sum_{i=0}^{n-1} (-1)^i \dim \operatorname{im} f_i \\ &= \sum_{i=1}^n (-1)^i \dim \ker f_i + \sum_{i=0}^{n-1} (-1)^i \dim \ker f_{i+1} \\ &= 0. \end{aligned}$$

Thus, the long exact sequence associated to  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$  gives us

$$\chi(\mathcal{F}') - \chi(\mathcal{F}) + \chi(\mathcal{F}'') = 0,$$

as desired. □

5.2. (a) Let  $X$  be a projective scheme over a field  $k$ , let  $\mathcal{O}_X(1)$  be a very ample invertible sheaf on  $X$  over  $k$ , and let  $\mathcal{F}$  be a coherent sheaf on  $X$ . Show that there is a polynomial  $P(z) \in \mathbb{Q}[z]$ , such that  $\chi(\mathcal{F}(n)) = P(n)$  for all  $n \in \mathbb{Z}$ . We call  $P$  the *Hilbert polynomial* of  $\mathcal{F}$  with respect to the sheaf  $\mathcal{O}_X(1)$ .

[*Hint:* Use induction on  $\dim \operatorname{Supp} \mathcal{F}$ , general properties of numerical polynomials (I, 7.3), and suitable exact sequences

$$0 \rightarrow \mathbb{R} \rightarrow \mathcal{F}(-1) \rightarrow \mathcal{F} \rightarrow \mathcal{D} \rightarrow 0.]$$

(b) Now let  $X = \mathbb{P}_k^r$ , and let  $M = \Gamma_*(\mathcal{F})$ , considered as a graded  $S = k[x_0, \dots, x_r]$ -module. Use (5.2) to show that the Hilbert polynomial of  $\mathcal{F}$  just defined is the same as the Hilbert polynomial of  $M$  defined in (I, 7).

*Proof.* (a)

(b) Let  $P(z)$  be the Hilbert polynomial of  $\mathcal{F}$  defined in (a). It suffices to show that  $P(n) = \dim M_n = \dim_k H^0(X, \mathcal{F}(n))$  for  $n \gg 0$ . However, (5.2) gives us

$$\begin{aligned} P(n) &= \chi(\mathcal{F}(n)) \\ &= \sum_{i=0}^n (-1)^i H^i(X, \mathcal{F}(n)) \\ &= H^0(X, \mathcal{F}(n)) \end{aligned}$$

for  $n \gg 0$ , so we're done.  $\square$

5.3. *Arithmetic Genus.* Let  $X$  be a projective scheme of dimension  $r$  over a field  $k$ . We define the *arithmetic genus*  $p_a$  of  $X$  by

$$p_a(X) = (-1)^r (\chi(\mathcal{O}_X) - 1).$$

Note that it depends only on  $X$ , not on any projective embedding.

(a) If  $X$  is integral, and  $k$  algebraically closed, show that  $H^0(X, \mathcal{O}_X) \cong k$ , so that

$$p_a(X) = \sum_{i=0}^{r-1} (-1)^i \dim_k H^{r-i}(X, \mathcal{O}_X).$$

In particular, if  $X$  is a curve, we have

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X).$$

[*Hint:* Use (I, 3.4).]

- (b) If  $X$  is a closed subvariety of  $\mathbb{P}_k^r$ , show that this  $p_a(X)$  coincides with the one defined in (I, Ex. 7.2), which apparently depended on the projective embedding.
- (c) If  $X$  is a nonsingular projective curve over an algebraically closed field  $k$ , show that  $p_a(X)$  is in fact a *birational* invariant. Conclude that a nonsingular plane curve of degree  $d \geq 3$  is not rational. (This gives another proof of (II, 8.20.3) where we used the geometric genus.)

*Proof.* (a) We have that  $H^0(X, \mathcal{O}_X) \cong k$  by (I, 3.4). Then

$$\begin{aligned} p_a(X) &= (-1)^r (\chi(\mathcal{O}_X) - 1) \\ &= (-1)^r (\chi(\mathcal{O}_X) - \dim_k H^0(X, \mathcal{O}_X)) \\ &= (-1)^r \left( \sum_{i=1}^r (-1)^i \dim_k H^i(X, \mathcal{O}_X) \right) \\ &= \sum_{i=1}^r (-1)^{r-i} \dim_k H^i(X, \mathcal{O}_X) \\ &= \sum_{i=0}^{r-1} (-1)^i \dim_k H^i(X, \mathcal{O}_X). \end{aligned}$$

If  $X$  is a curve, then  $r = 1$ , so the result holds.

- (b)
- (c)

$\square$

5.4. Recall from (II, Ex. 6.10) the definition of the Grothendieck group  $K(X)$  of a noetherian scheme  $X$ .

- (a) Let  $X$  be a projective scheme over a field  $k$ , and let  $\mathcal{O}_X(a)$  be a very ample invertible sheaf on  $X$ . Show that there is a (unique) additive homomorphism

$$P : K(X) \rightarrow \mathbb{Q}[z]$$

such that for each coherent sheaf  $\mathcal{F}$  on  $X$ ,  $P(\gamma(\mathcal{F}))$  is the Hilbert polynomial of  $\mathcal{F}$  (Ex. 5.2).

- (b) Now let  $X = \mathbb{P}_k^r$ . For each  $i = 0, 1, \dots, r$ , let  $L_i$  be a linear space of dimension  $i$  in  $X$ . Then show that

(i)  $K(X)$  is the free abelian group generated by  $\{\gamma(\mathcal{O}_{L_i}) \mid i = 0, \dots, r\}$ , and

(ii) the map  $P : K(X) \rightarrow \mathbb{Q}[z]$  is injective.

[Hint: Show that (i)  $\implies$  (ii). Then prove (i) and (ii) simultaneously, by induction on  $r$ , using (II, Ex. 6.10c).]

*Proof.* (a) The obvious map on the free abelian group generated by all coherent sheaves of  $X$  descends to  $K(X)$  by (Ex. 5.1) above.

- (b) If (i) holds, then  $P$  is injective because  $P(\gamma(\mathcal{O}_{L_i}))$  □

5.7. Let  $X, Y$  be proper schemes over a noetherian ring  $A$ . We denote by  $\mathcal{L}$  an invertible sheaf.

- (a) If  $\mathcal{L}$  is ample on  $X$ , and  $Y$  is any closed subscheme of  $X$ , then  $i^*\mathcal{L}$  is ample on  $Y$ , where  $i : Y \rightarrow X$  is the inclusion.

- (b)  $\mathcal{L}$  is ample on  $X$  if and only if  $\mathcal{L}_{\text{red}} = \mathcal{L} \otimes \mathcal{O}_{X_{\text{red}}}$  is ample on  $X_{\text{red}}$ .

- (c) Suppose  $X$  is reduced. Then  $\mathcal{L}$  is ample on  $X$  if and only if  $\mathcal{L} \otimes \mathcal{O}_{X_i}$  is ample on  $X_i$  for each irreducible component  $X_i$  of  $X$ .

- (d) Let  $f : X \rightarrow Y$  be a finite surjective morphism, and let  $\mathcal{L}$  be an invertible sheaf on  $Y$ . Then  $\mathcal{L}$  is ample on  $Y$  if and only if  $f^*\mathcal{L}$  is ample on  $X$ .

[Hints: Use (5.3) and compare (Ex. 3.1, Ex. 3.2, Ex. 4.1, Ex. 4.2).]

*Proof.* (a) Let  $\mathcal{F}$  be a coherent sheaf on  $Y$ .  $i^*$  is a left adjoint functor and tensor products are direct limits, so  $(i^*\mathcal{L})^n = i^*(\mathcal{L}^n)$ . Thus, we may omit the parentheses without ambiguity. By (II, Ex. 5.1d), we have

$$i_*(\mathcal{F} \otimes_{\mathcal{O}_Y} i^*\mathcal{L}^n) \cong i_*\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n.$$

$X, Y$  are noetherian and  $i$  is a closed immersion, so  $i_*\mathcal{F}$  is coherent by (II, Ex. 5.5). Then (5.3) gives us  $n_0 > 0$  such that, for all  $n > 0$  and  $p > 0$ ,

$$H^p(Y, \mathcal{F} \otimes i^*\mathcal{L}^n) \cong H^p(X, i_*\mathcal{F} \otimes \mathcal{L}^n) \cong 0$$

by (2.10). By (5.3) once more, we conclude that  $i^*\mathcal{L}$  is ample.

(b) We begin by noting that the immersion  $i : X_{\text{red}} \hookrightarrow X$  is a bijection, so  $i^*\mathcal{L} = \mathcal{L} \otimes \mathcal{O}_X = \mathcal{L}_{\text{red}}$ . If  $\mathcal{L}$  is ample on  $X$ , then  $\mathcal{L}_{\text{red}}$  is ample on  $X_{\text{red}}$  by (a). Suppose then that  $\mathcal{L}_{\text{red}}$  is ample on  $X_{\text{red}}$  and fix a coherent sheaf  $\mathcal{F}$  on  $X$ . As in (3.1), noetherianity of  $X$  gives a finite filtration  $\mathcal{F} \supset \mathcal{N}\mathcal{F} \supset \dots \supset \mathcal{N}^n\mathcal{F} = 0$ , and, thus, the exact sequences

$$0 \rightarrow \mathcal{N}^{k+1}\mathcal{F} \rightarrow \mathcal{N}^k\mathcal{F} \rightarrow \frac{\mathcal{N}^k\mathcal{F}}{\mathcal{N}^{k+1}\mathcal{F}} \rightarrow 0$$

for  $0 \leq k < n$ .

Note that the quotient sheaves are coherent by coherence of  $\mathcal{N}^k \mathcal{F}$  for all  $k \geq 0$ . Then

$$\mathcal{N} \subset \text{Ann} \left( \frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}} \right),$$

supplies us with sheaves  $\mathcal{G}_k$  on  $X_{\text{red}}$  such that  $i_* \mathcal{G}_k$ . In fact, the  $\mathcal{G}_k$  are coherent. This is because  $\text{im } \bar{f} = \text{im } (\bar{f} \circ \pi) = \text{im } f$  whenever  $\pi$  is surjective, so in particular when  $\pi : A \rightarrow A/\mathfrak{a}$  is the projection. Then (5.3) supplies  $n_0 > 0$  such that, for all  $n \geq n_0$  and  $p > 0$ ,  $H^p(X_{\text{red}}, \mathcal{G}_k \otimes \mathcal{L}^n) \cong 0$ .

Applying the projection formula (II, 5.1) gives us

$$H^\bullet(X, \frac{\mathcal{N}^k \mathcal{F}}{\mathcal{N}^{k+1} \mathcal{F}} \otimes \mathcal{L}^n) \cong H^\bullet(X_{\text{red}}, \mathcal{G}_k \otimes \mathcal{L}^n).$$

Tensoring our original exact sequence with  $\mathcal{L}^n$  for  $n \geq n_0$  and taking the long exact sequence gives us exactness of

$$H^p(X, \mathcal{N}^{k+1} \mathcal{F} \otimes \mathcal{L}^n) \rightarrow H^p(X, \mathcal{N}^k \mathcal{F} \otimes \mathcal{L}^n) \rightarrow H^p(X_{\text{red}}, \mathcal{G}_k \otimes \mathcal{L}^n) = 0$$

for  $p > 0$ .  $\mathcal{N}^n \mathcal{F} = 0$ , so it follows by induction on  $n - k$  that

$$H^p(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$$

for all  $p > 0$ , whence  $\mathcal{L}$  is ample by (5.3).

(c)  $\mathcal{L}_i := \mathcal{L} \otimes X_i$  is slightly abusive notation for the inverse image of the inclusion  $X_i \hookrightarrow X$ , where  $X_i$  is given the reduced induced subscheme structure. Thus, one direction is a consequence of (a). It suffices to show that  $\mathcal{L}$  is ample if  $\mathcal{L}_i$  is ample for each irreducible component  $X_i$  of  $X$ .

Let  $X = X_1 \cup \dots \cup X_r$  be an irreducible decomposition of  $X$ , finite since  $X$  is noetherian. Suppose that  $\mathcal{L}_i$  is ample for each  $i = 1, \dots, r$ , and consider the finite filtration  $\mathcal{F} \supset \mathcal{I}_{X_1} \mathcal{F} \supset \dots \supset \mathcal{I}_{X_1 \cup \dots \cup X_r} \mathcal{F} = 0$ , where each  $X_1 \cup \dots \cup X_k$  is given the reduced induced subscheme structure. We once again receive exact sequences

$$0 \rightarrow \mathcal{I}_{Y_1 \cup \dots \cup Y_{k+1}} \mathcal{F} \rightarrow \mathcal{I}_{Y_1 \cup \dots \cup Y_k} \mathcal{F} \rightarrow \frac{\mathcal{I}_{Y_1 \cup \dots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \dots \cup Y_{k+1}} \mathcal{F}} \rightarrow 0$$

for all  $0 \leq k < r$ , where

$$\mathcal{I}_{Y_{k+1}} \subset \text{Ann} \left( \frac{\mathcal{I}_{Y_1 \cup \dots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \dots \cup Y_{k+1}} \mathcal{F}} \right).$$

As above, we have coherent sheaves  $\mathcal{G}_{k+1}$  on  $Y_{k+1}$  such that

$$H^\bullet(X, \frac{\mathcal{I}_{Y_1 \cup \dots \cup Y_k} \mathcal{F}}{\mathcal{I}_{Y_1 \cup \dots \cup Y_{k+1}} \mathcal{F}}) \cong H^\bullet(Y_{k+1}, \mathcal{G}_{k+1}).$$

Then (5.3) supplies us with  $n_0 > 0$  such that

$$H^p(Y_{k+1}, \mathcal{G}_{k+1} \otimes \mathcal{L}_{k+1}^n) = 0$$

for all  $p > 0$  and  $n \geq n_0$ . Once again, this gives us surjectivity of the maps

$$H^p(X, \mathcal{I}_{Y_1 \cup \dots \cup Y_{k+1}} \mathcal{F}) \rightarrow H^p(X, \mathcal{I}_{Y_1 \cup \dots \cup Y_k} \mathcal{F})$$

whenever  $p > 0$ , so the result follows by induction on  $r - k$ .

(d) Suppose first that  $\mathcal{L}$  is ample on  $X$ . Let  $\mathcal{F}$  be a coherent sheaf on  $Y$ . By (II, Ex. 5.1d), we have

$$f_*(\mathcal{F} \otimes_{\mathcal{O}_Y} f^* \mathcal{L}^n) \cong f_* \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^n.$$

$X, Y$  are noetherian and  $f$  is a finite map, so  $f_*\mathcal{F}$  is coherent by (II, Ex. 5.5). Then (5.3) gives us  $n_0 > 0$  such that, for all  $n > 0$  and  $p > 0$ ,

$$H^p(Y, \mathcal{F} \otimes f^*\mathcal{L}^n) \cong H^p(X, f_*\mathcal{F} \otimes \mathcal{L}^n) \cong 0$$

by (2.10). By (5.3) once more, we conclude that  $f^*\mathcal{L}$  is ample.

Suppose now that  $f^*\mathcal{L}$  is ample. □

5.8. Prove that every one-dimensional proper scheme  $X$  over an algebraically closed field  $k$  is projective.

- (a) If  $X$  is irreducible and nonsingular, then  $X$  is projective by (II, 6.7).
- (b) If  $X$  is integral, let  $\tilde{X}$  be its normalization (II, Ex. 3.8). Show that  $\tilde{X}$  is complete and nonsingular, hence projective by (a).

Let  $f : \tilde{X} \rightarrow X$  be the projection. Let  $\mathcal{L}$  be a very ample invertible sheaf on  $\tilde{X}$ . Show there is an effective divisor  $D = \sum P_i$  on  $\tilde{X}$  with  $\mathcal{L}(D) \cong \mathcal{L}$ , and such that  $f(P_i)$  is a nonsingular point of  $X$ , for each  $i$ .

Conclude that there is an invertible sheaf  $\mathcal{L}_0$  on  $X$  with  $f^*\mathcal{L}_0 \cong \mathcal{L}$ . Then use (Ex. 5.7d), (II, 7.6), and (II, 5.16.1) to show that  $X$  is projective.

- (c) If  $X$  is reduced, but not necessarily irreducible, let  $X_1, \dots, X_r$  be the irreducible components of  $X$ . Use (Ex. 4.5) to show that  $\text{Pic } X \rightarrow \bigoplus \text{Pic } X_i$  is surjective. Then use (Ex. 5.7c) to show  $X$  is projective.
- (d) Finally, if  $X$  is any one dimensional proper scheme over  $k$ , use (2.7) and (Ex. 4.6) to show that  $\text{Pic } X \rightarrow \text{Pic } X_{\text{red}}$  is surjective. Then use (Ex. 5.7b) to show  $X$  is projective.

*Proof.* (a) This is exact (ii)  $\implies$  (i) of (II, 6.7).

(b) □